R16

Code No:131AG

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD
B .Tech I Year I Semester Examinations, December - 2018
ENGINEERING CHEMISTRY
(Common to EEE, ECE, CSE, EIE, IT)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a)	Why hardness is expressed in terms of calcium carbonate equivalent?	[2]
b)	Which salts are responsible for temporary and permanent hardness of water?	[3]
c)	Why do electrochemical cells stop working after some time?	[2]
d)	Write a short note on calomel electrode.	[3]
e)	What is the functionality of a monomer?	[2]
f)	What are the applications of Bakelite?	[3]
g)	What are primary and secondary fuels?	[2]
h)	What is CNG? Why it is preferred over LPG?	[3]
i)	How is thermal conductivity of a refractory related to its porosity?	[2]
j)	What are the characteristics of good refractories?	[3]

PART - B

(50 Marks)

2.a)	Discuss the ion exchange process of softening hard water.	
b)	What is Caustic embrittlement?	[8+2]
	OR	
3.a)	What is potable water? What are the steps taken to obtain pure drinking water?	
b)	Write the names of three sludge forming and three scale forming compounds.	[7+3]
4.a)	Describe the construction of lead - acid battery with the reactions occurring during discharge.	
b)	Write short note on glass electrode.	[6+4]
	OR	
5.a)	Distinguish between galvanic cell and a concentration cell.	
b)	Write a short note on factors influencing e.m.f of batteries.	[6+4]
6.a)	Explain addition and condensation polymerisation with suitable examples.	
b)	Write a short note on vulcanization process.	[7+3]
	OR	
7.a)	What is chain polymerization and explain the steps involved?	
b)	Classify conducting polymers with suitable example.	[6+4]

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8.a) Explain refining of petroleum.
b) Why a good fuel must have low ash content? [7+3]

OR

9.a) Explain ultimate analysis of coal.
b) Define calorific value of fuel. [8+2]

10.a) Explain manufacture of cement in detail.
b) Write any two types of refractories and their uses. [8+2]

OR

11.a) Discuss four essential properties of a good refractory material.
b) Write the classification of lubricants with example. [6+4]

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